



KU LEUVEN

Analyzing Cycling Traffic Resilience to Weather Conditions in Flanders

Group 7

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1. Introduction

1.1 Context of the study

Cycling has increasingly become an important mode of transportation. In Flanders, Agentschap Wegen en Verkeer (AWV), monitors the cycling traffic with a network of counting stations. The AWV data provides valuable insight into the usage of the cycling infrastructure in Flanders.

It is important to consider external factors, when analysing cycling behaviour, as these may also influence the cycling activity. This study will analyse weather conditions as a possible influence on cycling activity. In particular, weather conditions may affect cycling demand by influencing visibility, and travel conditions. It is therefore important to understand how environmental factors influence cycling behaviour, to interpret traffic patterns and for infrastructure planning and design considerations which may support cycling in various conditions.

The aim of this paper is to examine the relationship between weather conditions and cycling activity in Flanders. To analyse this relationship, the present study will combine the AWV dataset with a meteorological observation dataset from the Royal Meteorological Institute (RMI). The study will try to identify how cycling behaviour varies across temporal, weather, and spatial conditions. The findings of this study may support infrastructure planning and decision making with regards to mobility.

1.2 General research question

The present study aims to provide an answer to the following research question:

How do weather conditions influence cycling activity across Flanders, and to what extent do cycling locations differ in their resilience to adverse weather conditions?

2. Data preparation and limitations

2.1 Data preparation

The analysis of the present study focuses on a merged dataset created from merging bicycle traffic dataset from AWV and dataset of meteorological observations from Royal Meteorological Institute (RMI). The merged dataset makes it possible to closely examine how environmental conditions influence cycling activity across different locations in Flanders. Several preprocessing steps were taken before the analytical phase of the study, since the two datasets were collected independently and differed in both temporal and spatial structures.

The AWV dataset contained automatic bicycle counts recorded at fixed monitoring stations distributed across the Flemish cycling network. Each observation in the dataset contained the following information: counting location, traffic direction, start and end timestamp, number of detected users within a given interval. The initial dataset also contained information about other modes of transportation, nevertheless, these were disregarded as the research objective of the present study is cycling. The temporal resolution of the cycling data corresponds to 15-minute intervals.

The RMI dataset was used to account for environmental information when considering traffic observations. The following variables from the RMI dataset were chosen for the present study: precipitation, temperature, wind speed, wind gusts, humidity, atmospheric pressure, solar radiation, and sunshine duration. These variables were chosen as they are directly related to cycling comfort, visibility, and route conditions, which may influence cycling demand.

A spatial matching procedure was used, for merging the datasets, to connect each cycling counter to its nearest weather station. The nearest RMI station and the corresponding distance in kilometres were stored in the merged dataset for every site. This approach relies on the assumption that geographically close weather stations provide sufficiently representative local weather conditions for the associated cycling site. An important consideration for this approach is that it may not always capture localized weather effects perfectly, particularly for sites located further from the meteorological station. However, the relatively dense coverage of both cycling counters and RMI stations across Flanders makes nearest-station matching a practical and computationally efficient treatment for the purposes of regional-scale analysis.

Temporal alignment was used to synchronize both datasets, as they originated from different recording systems, for the merging procedure. All timestamps were converted to UTC datetime format. The merged timestamp variable corresponds to the end of each 15-minute cycling interval.

The final merged dataset contains both spatial and temporal dimensions of cycling activity together with synchronized weather information.

2.2 Limitations

Several limitations of the present analysis should be acknowledged.

First, weather conditions were attributed to cycling sites using a nearest-station matching approach. For sites located far from the nearest RMI station, the matched weather data may not accurately reflect local conditions, particularly for precipitation which is highly spatially variable. The high feature importance of `nearest_station_km` in the Random Forest model suggests this may have introduced noise into the weather-related predictors.

Second, the model does not include a holiday variable. Public and school holidays in Belgium significantly alter cycling patterns, particularly at leisure-oriented sites, and their absence may cause the model to misattribute holiday effects to weather or temporal variables.

Third, due to computational constraints, the Random Forest was trained on a 200,000-row subsample of the available training data. Training on the full dataset may further improve predictive performance, although the subsample already provided stable and interpretable results.

Fourth, the clustering solution produced a modest average silhouette score of 0.251, indicating moderate rather than strong separation between clusters. While the clusters remained interpretable, some overlap between site profiles likely exists.

Moreover, the analysis covers the period from August 2019 to March 2026, which includes the COVID-19 pandemic years. Unusual cycling patterns during 2020 and 2021 may affect the estimated relationships between weather and cycling demand, particularly for commuter-oriented sites where lockdown measures drastically reduced traffic volumes.

Finally, we have to acknowledge the potential risk of overfitting in the Random Forest model: the selected model is fit with a relatively high maximum tree depth of 20, which allows the model to capture complex non-linear relationships and interaction effects, but may also increase sensitivity to noise in the training data. This is reflected in the difference between the training and test performance: the model reached R^2 of 0.644 on the training set and 0.403 on the test set. While it does not look like extreme overfitting, it still means that part of the model's predictive performance may depend on patterns that are not fully generalizable to unseen observations. Simpler model specifications were also explored, such as a maximum depth of 10, which produced more similar training and test scores but substantially lower overall performance. Therefore, the final Random Forest model should be interpreted as a trade-off between predictive accuracy and model complexity, and its results should be read with caution.

3. Exploratory data analysis

The goal of the exploratory data analysis was to identify the main trends and patterns in cycling activity. Due to the scale of the merged dataset, a reproducible 5% random sample was used to reduce computational overhead while preserving the overall structure of the data.

After converting UTC timestamps to Brussels local time several temporal features including hour of day, month, and weekday indicators were extracted.

3.1 Temporal cycling patterns

The analysis revealed patterns of daily and seasonal nature with regards to cycling activity. The cycling traffic on weekday seems to follow a pronounced bimodal distribution, with peaks occurring during morning and evening commuting hours. The weekend traffic appears to be more evenly distributed throughout the day. Seasonal variation is also visible, with higher cycling volumes during spring and summer months and lower activity for autumn and winter.

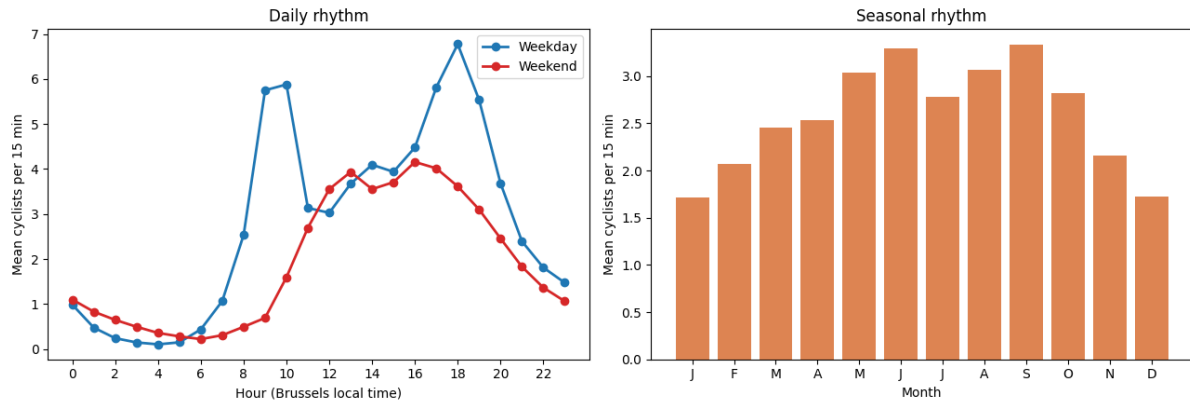


Figure 1: Average cycling volume by hour and day type and monthly variation in average cycling counts

These patterns indicate that temporal effects may influence cycling behavior and provide the rationale for incorporating them into predictive models.

3.2 Weather effects on cycling activity

In order to assess potential sensitivity of cycling behavior to weather conditions, cycling counts were compared across different rainfall conditions. The precipitation values were grouped into categorical intensity levels ranging from dry to heavy rainfall. The average cycling counts were then compared across these categories.

The results showed a decline in cycling activity as precipitation intensity increases. The dry intervals appeared to have substantially higher average counts as compared to moderate or heavy rainfall intervals. This pattern supports the hypothesis that adverse weather discourages cycling activity.

The interaction effect between rainfall and time of day was examined by comparing hourly cycling volumes during dry intervals and intervals with rainfall.

It can be seen from the figure that in intervals with rainfall cycling counts are lower throughout the day, nevertheless, the effect is not uniform across hours. It is also worth noting that commuting peaks were visible even during rainy periods. This suggests that weather conditions may influence cycling behavior more for recreational cyclists. This observation provides the rationale for the use of models capable of capturing interaction effects between weather conditions and temporal features.

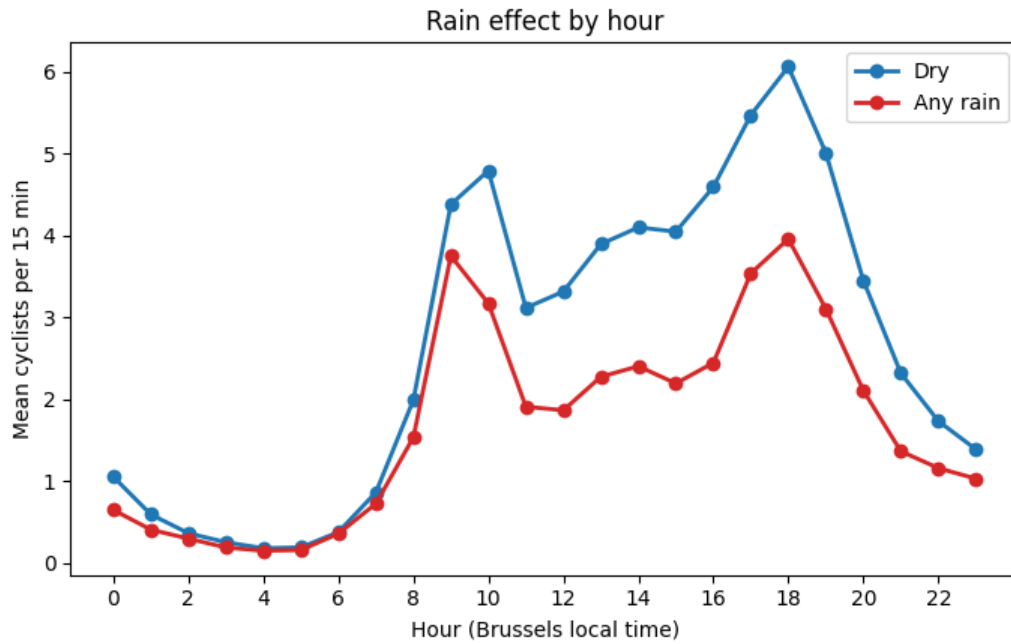


Figure 2: Hourly cycling profiles during dry and rainy conditions.

3.3 Spatial heterogeneity across sites

To quantify the variability that exists between cycling locations in both baseline traffic volume and weather sensitivity, average cycling counts were computed during dry periods for each site and compared to average counts during intervals with rainfall. The difference expressed in percentage between dry conditions and rainy conditions was then used as a proxy for rain sensitivity.

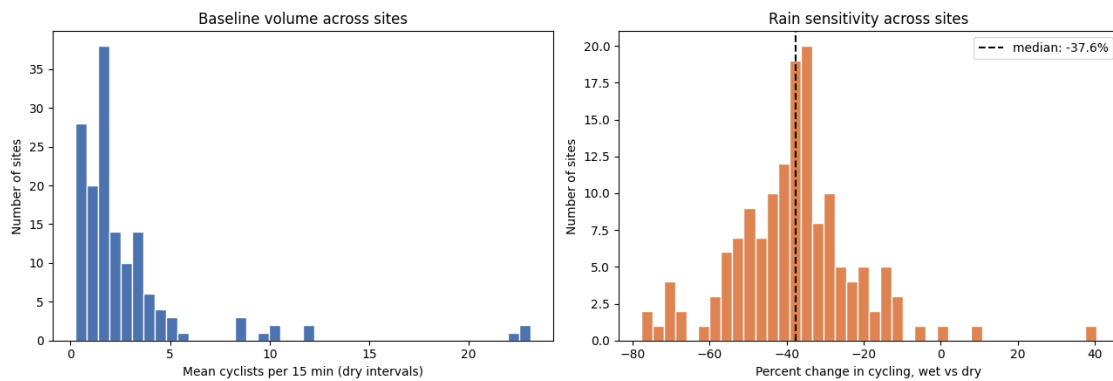


Figure 3: Distribution of baseline cycling volume and rain sensitivity across sites

The distributions reveal substantial heterogeneity between monitoring locations. Some sites maintain relatively stable traffic volumes during rainfall, on the other hand, some experience declines in cyclist counts. This variation suggests that cycling behavior is not only shaped by weather conditions, but also by spatial context such as commuting patterns or infrastructure characteristics.

Spatial variability was further examined geographically by mapping both the distance to the nearest weather station and the average cycling volume per site.

It can be seen from the figure that cycling intensity differs across regions, with higher intensity being in urban areas.

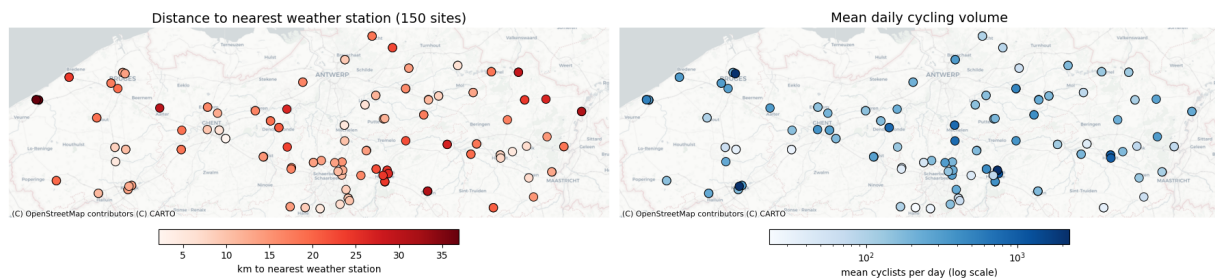


Figure 4: Weather-station proximity and cycling intensity across Flanders.

3.4 Relationships between weather variables

In order to examine relationships between weather variables and to identify potential multicollinearity issues before model estimation, a correlation analysis was done.

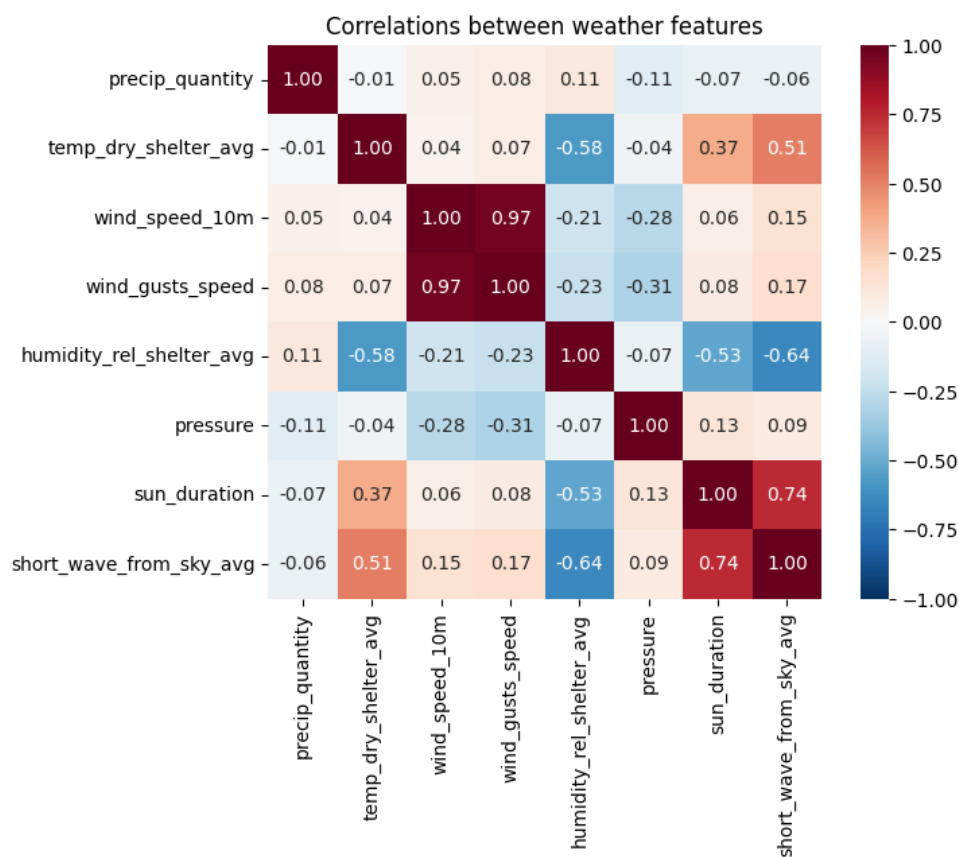


Figure 5: Correlation matrix of weather variables

Several relationships emerged from this analysis. Positive correlation was found between sunshine duration and solar radiation, and negative correlation between humidity and variables related to temperature.

3.5 Summary of EDA

Overall, the exploratory analysis suggests that cycling activity varies with temporal, weather, and site-specific conditions. These observations provide the rationale for the subsequent analyses.

4. Predictive models

4.1 Linear regression

In order to examine the relationship between cycling counts and weather-related variables, a Lasso regression model was used. This modelling approach was chosen for its interpretability and its ability to perform automatic variable selection. In the context of this study, the model allows for a direct estimation of how changes in weather conditions are associated with changes in cycling demand, while penalising less informative predictors toward zero.

The target variable is the log-transformed cyclist count recorded during each 15-minute interval. Log transformation was applied to reduce the influence of extreme values and to improve the distributional properties of the target. The predictor set included precipitation quantity, temperature, wind speed, wind gusts, humidity, solar radiation, hour of day, weekday, weekend indicator, month, commute hour indicator, rain dummy, heavy rain dummy, and distance to the nearest weather station. The dataset was split into training (80%) and test (20%) sets prior to any preprocessing to prevent data leakage. Missing values were imputed using column means and all numerical features were standardised within the pipeline, fitted on training data only.

The regularisation parameter alpha was selected via 5-fold cross-validation over the candidate values {0.001, 0.01, 0.1, 1.0, 10.0}. The optimal value was $\alpha = 0.001$, suggesting that minimal regularisation was sufficient and that most predictors contribute meaningfully to the model. The model achieved an R^2 of 0.165 on the test set with an MSE of 0.704.

The coefficient estimates reveal that temporal features are the dominant drivers of cycling demand. The commute hour indicator has the largest positive coefficient (0.169), followed by solar radiation (0.155) and hour of day (0.129). This indicates that time-of-day structure – particularly the morning and evening commuting peaks – explains more variation in cycling counts than weather conditions. Humidity shows the strongest negative association (-0.097), followed by weekend (-0.068) and wind gusts (-0.061). Notably, Lasso shrunk the weekday coefficient to exactly zero, removing it from the model entirely.

The direct rainfall variables have negligible coefficients: the rain dummy (-0.004), precipitation quantity (0.003), and heavy rain indicator (0.001) are all close to zero and rank at the bottom of the coefficient table. This does not mean that rain has no effect on cycling, but rather that once time-of-day and atmospheric conditions such as humidity and solar radiation are controlled for, the marginal contribution of the rainfall indicators is small at the global level. This finding is consistent with the hypothesis that weather effects on cycling may be heterogeneous across sites, which is explored further in the clustering analysis.

4.2 Random forest

The Random Forest model (100 trees, max depth 20, selected via 3-fold cross-validation across 4 candidate configurations) achieved an R^2 of 0.403 on the test set, explaining 40% of the variance in log-transformed cycling counts. This represents a substantial improvement over the Lasso baseline ($R^2 = 0.165$), indicating that non-linear relationships and interaction effects between weather and temporal variables are present in the data.

Feature importance rankings reveal that temporal and atmospheric conditions dominate cycling demand. Time-of-day (hour, 17.6%) and distance to the nearest weather station (nearest_station_km, 23.5%) are the two strongest predictors, followed by humidity (11.3%), temperature (10.6%), and solar radiation (9.8%). Direct rainfall measures rank consistently at the bottom: precip_quantity accounts for only 0.8% of importance, rain for 0.3%, and heavy_rain for less than 0.1%. Importantly, the high importance of nearest_station_km likely reflects a data quality effect (sites further from weather stations have less reliable weather measurements) and should be interpreted with caution rather than as a substantive finding.

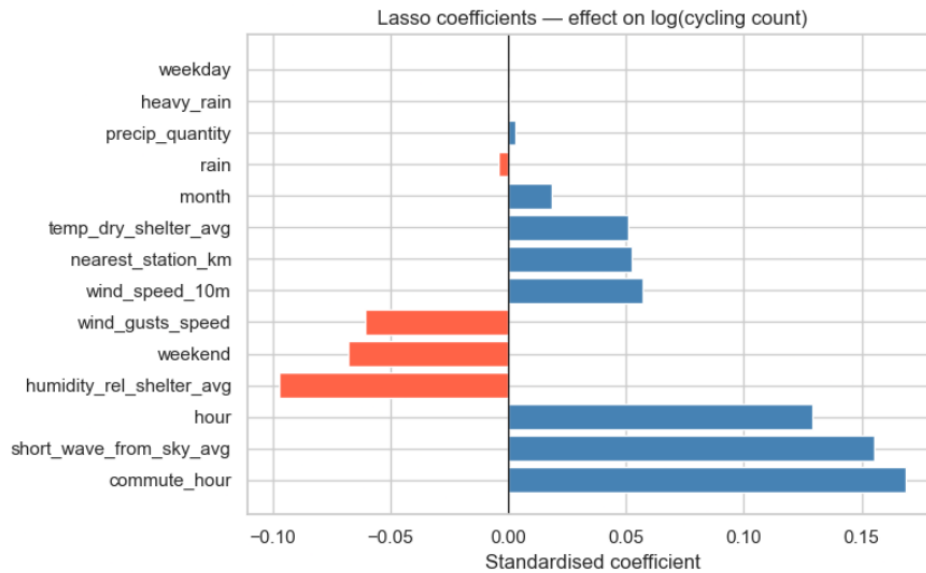


Figure 6: Lasso coefficients

The low importance of direct rainfall variables at the global level is itself an informative result. It suggests that, averaged across all 149 sites and the full 2019-2026 period, rainfall variables contribute less to predictive performance relative to temporal and broader atmospheric variables on cycling counts once time-of-day and broader atmospheric conditions are controlled for. However, this aggregate picture may mask substantial heterogeneity across sites, which motivates the subsequent clustering analysis. The site-level rain sensitivity scores derived from the model range from -0.008 to +0.003 (in log-count units), with a mean of -0.0006 and standard deviation of 0.002, indicating that while the average effect of rain is near zero, meaningful variation exists across locations. Sites with more negative sensitivity scores represent locations where rainfall is associated with larger drops in cycling traffic (the “weather-sensitive” sites), while sites near zero or positive represent more resilient locations.

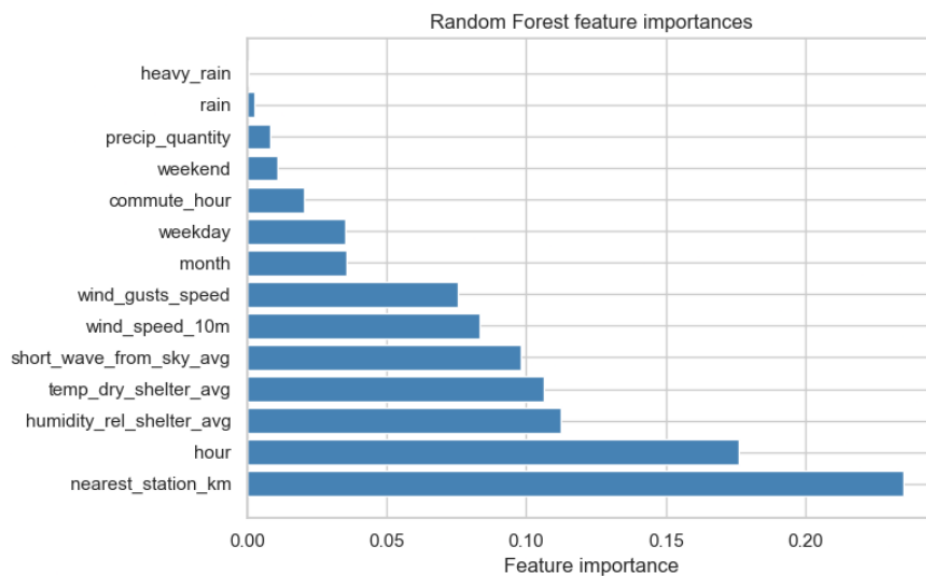


Figure 7: Random Forest feature importances

5. Clustering

5.1 K-means fit

Clustering analysis was performed to explore whether cycling sites can be grouped according to similar weather sensitivity profiles; it focuses on the heterogeneity between sites specifically to identify if some groups of sites are more resilient or more sensitive to weather conditions than others.

For this purpose, a site-level dataset was created, where each observation in the dataset corresponds to one cycling site. For each site, we used variables that describe both its usual cycling volume and its sensitivity to weather. Main clustering variables were the site-level coefficients for rain, precipitation quantity, heavy rain, temperature, wind gusts, and weekends. These coefficients indicate whether cycling counts at a given site tend to decrease or increase under these conditions. For example, a more negative rain coefficient means that the site is more sensitive to rainfall. We also included the median daily cycling count to account for the typical traffic level of the site. Finally, of course, we included the share of rainy days and the share of heavy-rain days, so that the clustering also accounts for how often each site was exposed to rainy conditions.

During this preparation step, wind speed at 10 meters was excluded from the final clustering variables, as the wind day was structurally missing for 25 out of 148 sites and retaining it would give missing data patterns too much influence.

Before clustering, the variables were standardized, since some variables had different scales, such as coefficients, rain shares, and cycling volumes. K-means is an algorithm based on Euclidean-distances, so cluster centroids' calculation on each iteration depends heavily on how spread each variable is; standardization eliminates the problem of bigger-scale variables automatically having more influence.

For the reasons mentioned above, K-means is also sensitive to outliers. We also removed some of them before estimating the final clusters. We used IsolationForest to identify the most unusual site profiles, and excluded them. The outlier threshold was set to 8%, which resulted in 12 out of 148 sites getting eliminated; remaining 136 sites were then used for the final clustering analysis.

Principal Component Analysis was then applied before the final k-means clustering, as several weather sensitivity variables may contain overlapping information; PCA therefore helps summarize the main dimensions of variation. We retained the first five principal components, which explained 81.9% of the variance in the standardized clustering variables. K-means was then applied to those five principal components.

We compared several values of k using inertia, silhouette scores, and cluster sizes, and retained the three-cluster solution as the best one. Even though it produced an average silhouette score of only 0.251, which implies modest separation between clusters, it produced interpretable and reasonably sized site profiles. The final solution contained 11 sites in Cluster 1, 65 sites in Cluster 2, and 60 sites in Cluster 3.

5.2 Clusters interpretation

We interpret Cluster 1 as a group of rain-exposed but relatively rain-resilient sites.

This cluster has an average rain coefficient of 0.001, which is very close to zero. This means that simple presence of rain is not associated with a strong decrease in cycling activity in this group. However, the precipitation quantity coefficient is still negative at -0.102, therefore larger amounts of rainfall still reduce cycling counts. This cluster also has the highest rain share at 0.671 and the lowest heavy-rain share at 0.125. This suggests that these sites often experience rainy conditions, but these conditions are more often mild rather than heavy rainfall. The median daily count in this cluster is 404.8 cyclists. The average distance to the nearest weather station is also the highest in this cluster, 20.5 km, so we interpret this cluster with a pinch of salt.

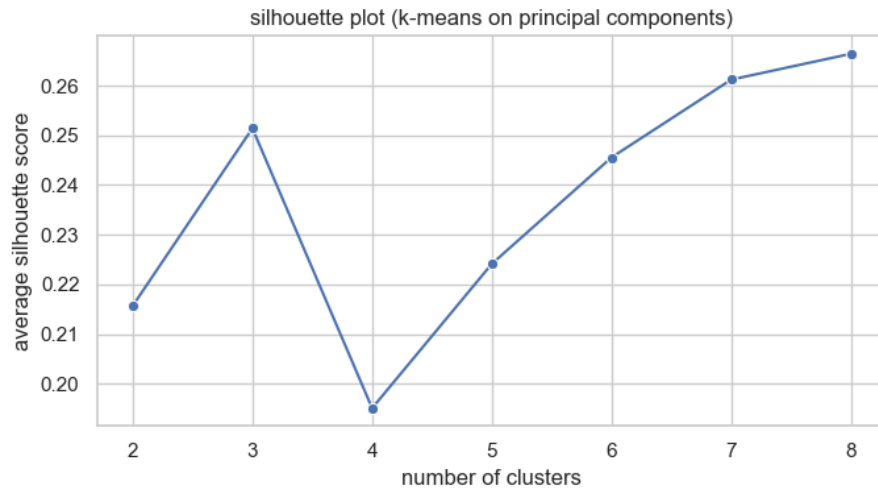


Figure 8: Silhouette plot for K-means on the first 5 principal components retained from the PCA.

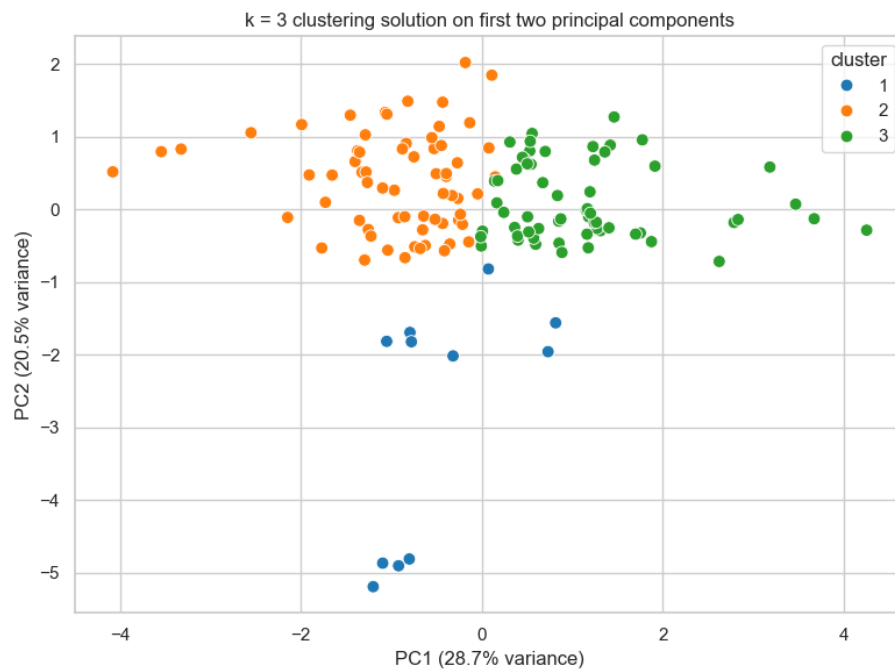


Figure 9: Final k-means solution visualized on the first two principal components.

We interpret Cluster 2 as a high-volume, commuter-like, and relatively weather-resilient group.

This cluster has the highest median daily cycling count of 582.7 cyclists. It also has the most negative weekend coefficient at -0.313, which suggests that cycling activity is stronger on weekdays than on weekends. This pattern looks commuting-adjacent. The precipitation coefficient is the least negative of the three clusters at -0.048, and the temperature coefficient is also the lowest at 0.153. This means that cycling activity in this group depends less on warmer temperatures and decreases less strongly with rainfall quantity. These sites therefore appear to represent routes with more stable demand.

We interpret Cluster 3 as a lower-volume and fairly weather-sensitive group.

This cluster has the lowest median daily cycling count of 292.5 cyclists. It also has the strongest negative rain coefficient at -0.076, and the strongest negative precipitation coefficient at -0.129. This indicates that cycling activity at these sites decreases more strongly when rain is present and when rainfall quantity increases. At the same time, this cluster has the highest temperature coefficient, at 0.296, meaning that cycling activity increases more strongly on warmer days. This combination suggests a relatively weather-dependent nature of a cluster, where cycling activity is more likely to increase in nice weather and decrease in bad weather. The weekend coefficient is less negative than in Cluster 2 at -0.139, which also means that these sites are less commuter-oriented and may represent more leisure-like cycling routes.

From a business or planning perspective, the clusters show that cycling sites with similar traffic volumes may still differ in weather resilience. High-volume commuter-like sites appear relatively robust and may be treated as core cycling corridors. Lower-volume fairly weather sensitive sites may need more targeted interventions, such as better surface quality, lighting, wind/rain protection, or improved route continuity. The clustering therefore helps in identifying where cycling demand is stable and where it is more vulnerable to adverse weather.

6. Conclusion

The results of our analysis suggest that cycling demand in Flanders is shaped primarily by regular mobility patterns rather than by weather conditions alone.

Across both the Lasso and Random Forest models, variables related to time-of-day structure, seasonality, humidity, temperature, and solar radiation consistently explained more variation in cycling counts than precipitation indicators. While exploratory analysis suggested a negative relationship between rainfall and cycling activity, the predictive models showed that rainfall variables contribute relatively little additional explanatory power once temporal and broader weather conditions are controlled for. This indicates that rainfall is not the dominant global predictor of cycling demand across all sites, although its effects may differ substantially between locations.

The choice of Lasso to handle multicollinear weather predictors and Random Forest to capture non-linear interaction effects was validated by the substantial improvement in R^2 from 0.165 to 0.403, confirming that weather and temporal variables interact in ways a linear model alone cannot represent.

We explored the previously mentioned heterogeneity with clustering. High-volume weather-robust sites have a relatively stable demand: they have the highest median daily counts, strong weekday orientation, and weaker penalties for precipitation quantity. They function more or less like commuting routes, so we would suggest that the planning priority in maintaining surface quality, safety, and reliability should be given to these routes. Weather-sensitive sites have the opposite pattern: lower median counts, stronger negative rain and precipitation coefficients, and a stronger positive temperature coefficient. Cycling at these locations appears more dependent on nicer conditions, so targeted improvements such as lighting, smoother surfaces, and protection from wind or rain could reduce weather-related drops in demand.

We believe that for AWW planners, maintenance teams, and the mobility policymakers, our analysis provides many interesting weather-related insights that are not otherwise visible in simple demand analysis; weather-resilience profiles allow us to identify how dependable demand is under adverse conditions. Such two-dataset analysis helps with infrastructure decisions for which it is necessary to distinguish between sites that already support stable daily cycling and sites where specific design improvements could make cycling less sensitive to bad weather.

In terms of the presentation of our results, we provide a map with sites colored by cluster, supported by a table with a median daily volume, rain sensitivity, and temperature sensitivity. This will help our target audiences and decision-makers distinguish between core cycling network assets (high-volume, weather-resilient) and more weather-sensitive sites that may be due for specific infrastructure improvements.